MIL-C-3098/109B
23 June 1975
SUPERSEDING
MIL-C-3098/109A
21 January 1972

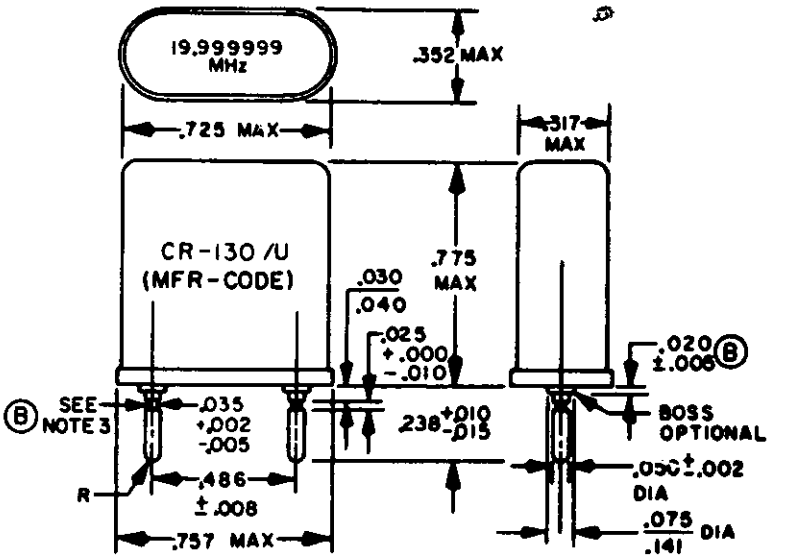
MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-130/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 0.8 to 20 MHz; fundamental; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.050	1.27
.005	.13	.075	1.91
.008	.20	.141	3.58
.010	.25	.238	6.05
.015	.38	.317	8.05
.020	.51	.352	8.94
.025	.64	.486	12.34
.030	.76	.725	18.42
.035	.89	.757	19.23
.040	1.02	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
(B) 3. The pin undercut may be omitted.

(B) denotes change

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TABLE I. Requirements.

Requirements	Specified value or condition	
Holder	HC-6/U (see MIL-H-10056/2)	
Dimensions, marking, and configuration	See figure	
Frequency range	0.8 to 20 MHz, inclusive	
Capacitance, shunt	7 pF, maximum	
Frequency tolerances (operating temperature range):		
Primary	±20 PPM	
Secondary	±30 PPM	
Equivalent resistance	See table II	
Antiresonance load capacitance	30 ±0.5 pF	
Mode of oscillation	Fundamental	
Temperature ranges:		
Primary	-40 to +90° C, inclusive	
Secondary	-55 to -40° C and +90 to +105° C, inclusive	
Test set, calibration values, and rated drive level	See table III	
Shock (specified pulse):	Below 2 MHz	2 MHz and above
Frequency change permitted	±10 PPM	±5 PPM
Equivalent resistance change permitted	±15 percent	±10 percent
Vibration, high frequency: 1/		
Frequency change permitted	±10 PPM	±5 PPM
Equivalent resistance change permitted	±15 percent	±10 percent
Temperature run:		
Frequency change permitted	±10 PPM	±5 PPM
Equivalent resistance change permitted	±15 percent	±10 percent
Aging:		
Frequency change permitted	5 PPM	5 PPM

1/ Method 204 of MIL-STD-202, test condition A.

TABLE II. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.8 to 0.85	620
0.85+ to 0.9	600
0.9+ to 1	570
1+ to 1.12	540
1.12+ to 1.25	490
1.25+ to 1.37	450
1.37+ to 1.5	410
1.5+ to 1.62	370
1.62+ to 1.75	330
1.75+ to 1.87	300
1.87+ to 2	290
2+ to 2.12	270
2.12+ to 2.25	240
2.25+ to 2.6	190
2.6+ to 3	150
3+ to 3.4	110
3.4+ to 3.75	90
3.75+ to 4	75
4+ to 5	60
5+ to 7	35
7+ to 10	24
10+ to 15	22
15+ to 20	20

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TABLE III. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	volts	mW
TS-330/TSM	0.8 to 1.5	100	10	---	10.0 ±2.0
	1.5+ to 2.25	50	15	---	10.0 ±2.0
	2.25+ to 3.4	40	15	---	10.0 ±2.0
	3.4+ to 5.1	25	20	---	10.0 ±2.0
	5.1+ to 7.5	14	25	---	10.0 ±2.0
	7.5+ to 10	11	30	---	10.0 ±2.0
	10+ to 15	13	20	---	5.0 ±1.0
TS-683/TSM	15+ to 20	10	--	0.22	5.0 ±1.0

NOTES:
1. The socket designed to mate with crystal holder HC-6/U is designated in MIL-S-12883/4 (Sockets, Crystal, Top and Bottom Mounting, 2 Contacts). For Navy use, other sockets listed in MIL-STD-242 may be used.
2. This crystal unit is similar to Crystal Type CR-85/U, except for operation at antiresonance and maximum equivalent resistance.

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army - MI, MU
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-0408-45)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-3098/109B CRYSTAL UNITS, QUARTZ, CR-130/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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